

Development of a Wheelset Roller Rig for Improving Rolling Stock Traction Efficiency

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ABSTRACT

This study presents the development of a novel wheelset roller rig designed to enable accurate and repeatable evaluation of rolling stock dynamics. Specifically, the proposed design aims to provide an experimental apparatus to better understand wheelset traction and curving dynamics. Ultimately, it supports the development of safe engineering practices that can improve the railroad's traction efficiency. The new system is designed to be assembled onto the existing Virginia Tech-Federal Railroad Administration (VT-FRA) Roller Rig. The VT-FRA Roller Rig has been used extensively in the past decade to understand the complex contact mechanics of the wheel-rail interface using a single wheel-roller setup.

The redesigned system introduces a second roller and a second wheel into the existing rig. The new roller operates independently, featuring its own motor-gear system and additional degrees of freedom (DoFs) that allow it to replicate field conditions more closely. Equipped with high-resolution actuators and a force measurement system, the new setup can replicate traction and braking under various creepage conditions. It also allows the emulation of key curving parameters, including angles of attack up to 3°, multiple lateral displacements, and cant angles up to 3°, while enabling precise measurement of the resulting contact forces. This study also discusses the gap between the curving dynamics observed in the field and the equivalent dynamics replicated in the roller rig.

Modeling of longitudinal creep forces confirms that the selected hardware meets the torque requirements and maintains velocity control within ± 0.0012 rpm sufficient to emulate curves with an error margin under ± 0.011 -degree curve. Despite a minor contact patch distortion due to roller diameter asymmetry, the system is highly effective for controlled testing of wheelset-rail interactions under various dynamic conditions.

INTRODUCTION

The contact forces developed at the wheel-rail interface are fundamentally important in determining railway vehicles' dynamic behavior, performance, and safety. These forces governed by the principles of contact

mechanics are responsible for how loads, vibrations, and stresses are transferred between the train and the track [1]. Beyond being one of the most challenging areas in railroad engineering, contact mechanics is critical for advancing understanding of rail vehicle behavior under real operating conditions. This field of research has led to key insights into modeling and mitigating wear, managing energy losses due to slippage, and preventing derailments. By studying the physics of wheel-rail interaction, researchers can propose more efficient traction and braking strategies and define safe operating zones for rolling stock.

For several decades, roller rigs have proven to be an effective method for studying complex contact mechanics. Unlike field conditions, roller rigs offer a controlled environment for evaluating various aspects of the wheel-rail interaction (WRI). They have been used to validate theories and software related to contact mechanics, assess vehicle stability under track irregularities, and evaluate rolling stock components [2]. There are multiple configurations of roller rigs used to replicate wheel-rail interactions. Some systems consist of a wheel in contact with a short section of rail, where the reciprocating motion of the rail simulates continuous wheel rotation over the track [3]. However, in most roller rigs, the rail is emulated by a rotating roller in contact with a rotating wheel, which is kinematically equivalent to a wheel rolling over a fixed rail.

Although roller rigs have consistently improved due to advancements in hardware and software technologies, further enhancements are still remarkably desired in experimental testing research. In the setup used by Matsumoto et al. [4], a scaled wheelset and two independently driven rollers were employed to investigate the characteristics of creep forces. While the creepages at the wheel-roller interface could be measured, there is no active control of the creep forces since the wheelset is idle. Additionally, the study does not discuss in depth how the curve negotiation is performed in the rig. Liu et al. performed experiments in a full-scale roller rig, providing alternatives to mimic the centrifugal forces in the roller rig [5]. The results demonstrated that a complex mechanical powertrain could achieve the necessary level of speed control for curve emulation. However, due to the size of the components, the full-scale rig reduced the accuracy of the creep force measurements, and the rig is intended for bogie dynamics studies only. The full-scale wheelset rig used by Bruni et al. could not achieve the speed differential in the roller unit since the rollers are mounted on the same shaft [6]. Also, the rig does not allow independent measurements of the contact forces, relying on the actuator's measurements. Kalousek et al. developed a 1/8th-scale rig for predominantly perform wear studies [7]. The significant difference in diameters between the wheel and the rollers causes contact patch distortions, and the rig lacked independent control of slip between the rotating bodies, limiting the rig's use for studies other than wear.

The review of multiple roller rigs highlights potential research gaps that still exist in the study of curving dynamics using wheelset roller rigs. No roller rig is currently equipped with a wheelset that can actively control creepage with high accuracy, as the wheels rotate idle (i.e., they cannot generate traction and break against the rollers). Therefore, there is a lack of experimental devices that can replicate wheelset traction or braking. Furthermore, to the best of the authors' knowledge, no existing studies have examined wheelset curving dynamics on roller rig systems or evaluated the level of accuracy required for operational control and realistic curve emulation.

Based on the identified gaps and potential opportunities, this study aims to enhance the Virginia Tech-Federal Railroad Administration (VT-FRA) Roller Rig capabilities to enable single axle/wheelset testing. The proposed improvements seek to provide a novel experimental setup to advance experimental traction and curving dynamics studies. The new configuration intends to replicate wheelset dynamics by introducing an independent roller and a scaled wheelset, adhering to the design guidelines:

- Accuracy: The design should maintain the same level of precision as the existing VT-FRA Roller Rig in terms of percentage creepage control and force measurement accuracy (0.01% and 2% Full Scale Output (FSO), respectively [8]) at both contact points.

- Longitudinal creepage control: Enabling consistent control of the longitudinal creepage for both wheels independently at the two contact points.
- Curving capabilities: Enabling a wide variety of curve radii emulation.
- Operation: Design the new mechanism with a modular structure to allow easy removal and reinstallation, enabling a straightforward return to the original system configuration if required

CURRENT SINGLE WHEEL-ROLLER VT-FRA ROLLER RIG

The Virginia Tech–Federal Railroad Administration (VT-FRA) Roller Rig is a state-of-the-art system designed to study single wheel-rail interactions with high accuracy. Its development involved extensive design evaluations, assessing multiple concepts before achieving the final design illustrated in Figure 1[8]. As depicted in the cross-sectional view shown in Figure 1b, the rig consists of two main elements: the single wheel on top and a large-diameter roller at the bottom. The ratio between the two is $1/5^{\text{th}}$. Wheels with different profiles can be interchanged on the wheel shaft depending on the test type (e.g., AAR-1B, AAR-2A, cylindrical wheel) while maintaining the $1/5$ ratio concerning the roller diameter. The roller profile mimics a standard AREMA RE136 rail that cannot be replaced for calibration and practical reasons. The linear scale of both rotating bodies are scaled by four (e.g., 12 inches in the field becomes 3 inches in the VT-FRA roller rig).

All components beyond the wheel and roller in the VT-FRA Roller Rig are designed exclusively to control the velocities and positions of these two bodies. As a result, the rig can replicate all key degrees of freedom (DoF) and creepages – relative slips between the wheel and rail – observed in real-world railway operations. Figure 1a highlights the four DoFs following the coordinate system. The vertical displacement along the Z-axis governs the wheel load and is controlled by two synchronized ball screw linear actuators. Designed for heavy-duty applications, these actuators offer low backlash and positional accuracy as fine as 0.05 mm. Together, they can apply a vertical load of up to 10 kN, corresponding to approximately 160 kN ($\approx 36,000$ lb.) in the field, based on the rig's force scaling factor [9]. The cant angle (rotation around the X-axis), a critical parameter for curve negotiation, is also replicated. Two additional linear actuators precisely control this motion, enabling wheel cant angles up to 3° . The angle of attack (rotation around the Z-axis) is emulated by yawing the roller. The roller is mounted on a rigid frame supported by a low-backlash cross roller ring bearing, and an additional linear actuator drives its yaw motion, allowing up to 5° of angle of attack—exceeding typical field conditions. Finally, lateral displacement along the Y-axis is implemented by shifting the roller relative to the wheel. This adjustment enables the roller to contact different regions of the wheel profile, including the tread and flange.

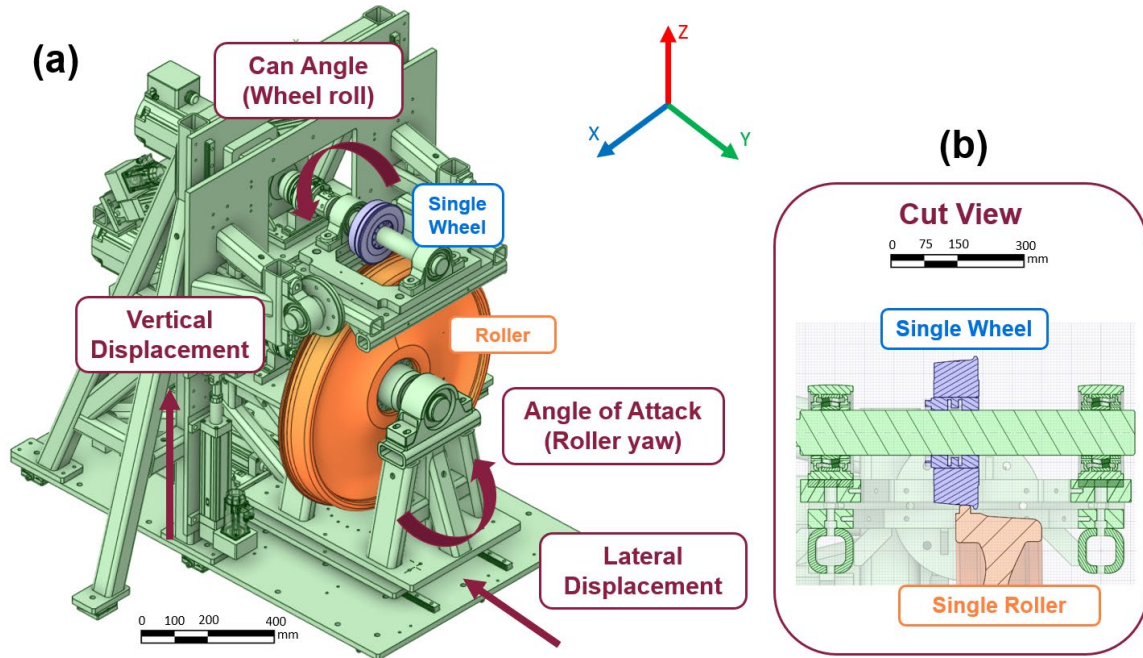


Figure 1: Current VT-FRA Roller Rig setup. The left side displays the four degrees of freedom (a), and the cross-sectional view on the right side (b) shows the single wheel and roller.

The precise control of the wheel and roller velocities (which creates the longitudinal creepages at the interface) is controlled by two independent electric motors. Each servomotor provides 15.7 kW of constant power and features a sine encoder that gives feedback to the closed-loop control with high digital resolution. This allows the wheel and roller to maintain velocity accuracies as low as 0.001 rotations per second (RPS) [10]. Such intricate design strategies enable the rig to conduct experiments with multiple creepages, thus enabling high fidelity in traction and braking assessments.

Additionally, the VT-FRA Roller Rig features a novel force measurement system designed to accurately measure the forces developed at the wheel-rail interface. The system comprises two dedicated platforms: the wheel and motor dynamometer. Each platform is equipped with four triaxial piezoelectric force sensors, which are statically calibrated. These sensors can measure both quasi-static and dynamic forces at the wheel-roller interface with high resolution. The eight triaxial sensors provide 24 independent force data channels sampled at 2000 Hz. The raw measurements are filtered and processed through a custom algorithm developed for the rig. This algorithm calculates the creep forces and the resulting moments at the wheel roller contact patch. Hence, the VT-FRA Roller Rig allows for high-fidelity evaluation of the complex force interactions during traction and curving tests in a single wheel-roller interface.

CURVING DYNAMICS: FIELD AND ROLLER RIG PERSPECTIVES

In conventional railway systems, the wheelset comprises two wheels mounted on a rigid axle. Each wheel has a conical shape with a flange on its inner edge. This conical geometry and wheel coupling are crucial for vehicle guidance. When a railcar moves along a straight track, the wheelset keeps the vehicle centered[2]. In the event of lateral disturbance, the difference in rolling radius created by the conical shape induces self-centering behavior. Since both wheels rotate at the same angular velocity, this generates a restorative motion that guides the wheelset back to the center of the track. This dynamic, first described by Klingel [11], is typically characterized by low-amplitude lateral oscillation (kinematic oscillation). Flange contact occurs for larger lateral displacements (usually beyond 7-10 mm in practical applications), creating a substantial lateral force called the gravitational stiffness effect. In straight-track conditions, wheel conicity

serves as the primary source of guidance. However, in sharper curves, flange contact becomes more significant.

When negotiating a curve, the high-rail wheel must travel a longer distance than the low-rail wheel. Because both wheels are fixed to the same axle and rotate at the same rate, the conical profiles naturally cause the wheelset to yaw and shift laterally to accommodate the different travel distances. This behavior guides the vehicle on the curve. Redtenbacher's formula first describes this phenomenon using a kinematic model that calculates the required lateral displacement (y) as a function of the nominal rolling radius (r_0), contact patch distance (l), curve radius (R) and the wheel conicity (λ) as shown in Eqn. (1) [12].

$$y = \frac{r_0 l}{R \lambda} \quad (1)$$

Although early models described only the kinematics of wheelsets, real-world dynamics are more complex. In a curve, lateral acceleration is generated, and the centrifugal forces tend to overturn the rail vehicle. To counteract this, tracks have superelevation, where the high rail is raised relative to the low rail, similar to roadways in turns, to reduce unbalanced lateral loads.

In addition to inertial effects, the relative slip between the wheel and the rail, known as creepage, plays a critical role in generating forces and moments at the wheel-rail interface. Due to the constraints imposed by the rigid axle and the vehicle's suspension system, creepage inevitably occurs in the longitudinal, lateral, and spin directions. These relative motions give rise to creep forces and moments, significantly influencing the vehicle's ability to safely and efficiently negotiate curves. The magnitude and direction of these forces are highly sensitive to several geometric parameters, most notably the angle of attack (yaw), lateral displacement, and cant angle. To accurately replicate curving dynamics in an experimental setup, these parameters must be controlled carefully.

In roller rigs where the rail is replicated by the rotating roller, the test setup must satisfy three fundamental requirements to replicate curving conditions, as illustrated in Figure 2 [1]. First, the rollers must be positioned radially, meaning they should be symmetrically aligned along an imaginary curve radius R (Figure 2a). This arrangement ensures the wheelset maintains the same relative position as it would on an actual curve. Second, the rollers must rotate at different speeds as shown in Figure 2b. For instance, in a left-hand curve, the right roller must rotate faster than the left roller, reflecting the real-world condition where the right-side wheel travels a longer path. The third requirement is that the roller unit must be able to incline to replicate the superelevation found in actual track curves.

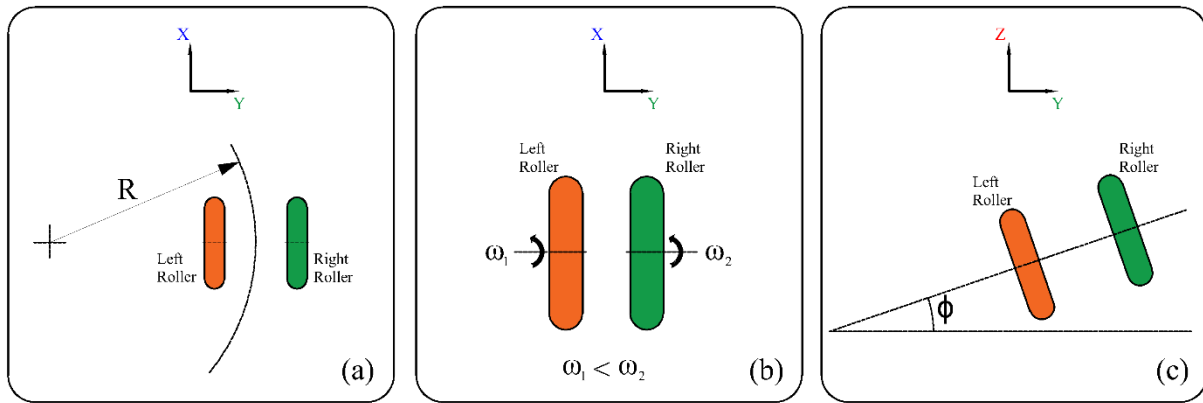


Figure 2: Requirements for curving emulation in a Roller Rig [1]. a) Set roller in the radial position; b) Left and Right roller with different angular velocities, and c) Roller unit tilting

As a roller-type system, the redesigned VT-FRA Roller Rig must meet these key requirements for curve emulation to accurately replicate field-like curving dynamics. To replicate the wheelset in the current setup, a second wheel is proposed to be installed on the rigid shaft. The existing roller (illustrated in Figure 1) will replicate the left rail (low rail in curving), and a new roller is proposed to replicate the right rail (high rail). The design effort and hardware selection to implement the wheelset and the new roller are discussed in Section 4.

The proposed configuration, however, presents several challenges for emulating curving dynamics. Under field conditions, the wheelset can freely yaw, roll, and displace laterally relative to the rails. The only constraint imposed on the wheelset is due to the vehicle's suspension system (i.e., springs and dampers in the bogie). In contrast, the proposed experimental setup relies on actuators to control the wheelset's positions relative to the rollers. A second challenge arises from the precise control of the roller speeds. Most curves in revenue service tracks are in the range of 1 to 4 degrees (i.e., have a large radii) to allow traveling at higher speeds and minimize lateral forces. Consequently, the difference in creepage and speed between the high- and low-rail wheels is minimal, barely distinguishable from tangent-track conditions. In the rig, replicating these subtle velocity differences demands high-precision control systems, increasing hardware complexity. A third challenge entails independently measuring creep forces at low and high contact patches. Vertical, lateral, and longitudinal forces are coupled since the wheels are mounted on a common rigid axle [13]. Accurately isolating and measuring these forces requires advanced sensor systems and force decoupling methods. Additionally, the wheelset experiences centrifugal forces resulting from vehicle motion in real curves, and the lateral acceleration cannot be directly replicated in the roller rig. This study assesses those challenges, and the outcome design intends to overcome them to enable proper curve emulation in the experimental setting.

VT-FRA ROLLER RIG ENHANCEMENTS

Concept evaluation

To enable wheelset traction experiments and assess the challenges of curving dynamics in experimental testing, the current VT-FRA Roller Rig has been adapted to accommodate an additional wheel and a second roller. However, the existing rig layout presents tight geometric constraints, which limit the size and placement of any new components. These constraints affect the frame dimensions, restrict the diameter of the second roller, and reduce flexibility in selecting actuators and motors for the new setup.

Given those constraints, multiple conceptual designs were developed and investigated. Figure 3 summarizes the wheelset and roller configurations, while TABLE 1 presents each concept's key characteristics, advantages, and limitations. In all six concepts illustrated in Figure 3, the left and right rollers are not symmetrical with respect to the wheelset, unlike traditional roller rigs. This asymmetry is the primary geometric constraint imposed by the existing rig. As a result, in concepts 1 through 4, the right roller has a significantly smaller diameter than the left. The remaining concepts explore alternative roller configurations to accommodate the available space.

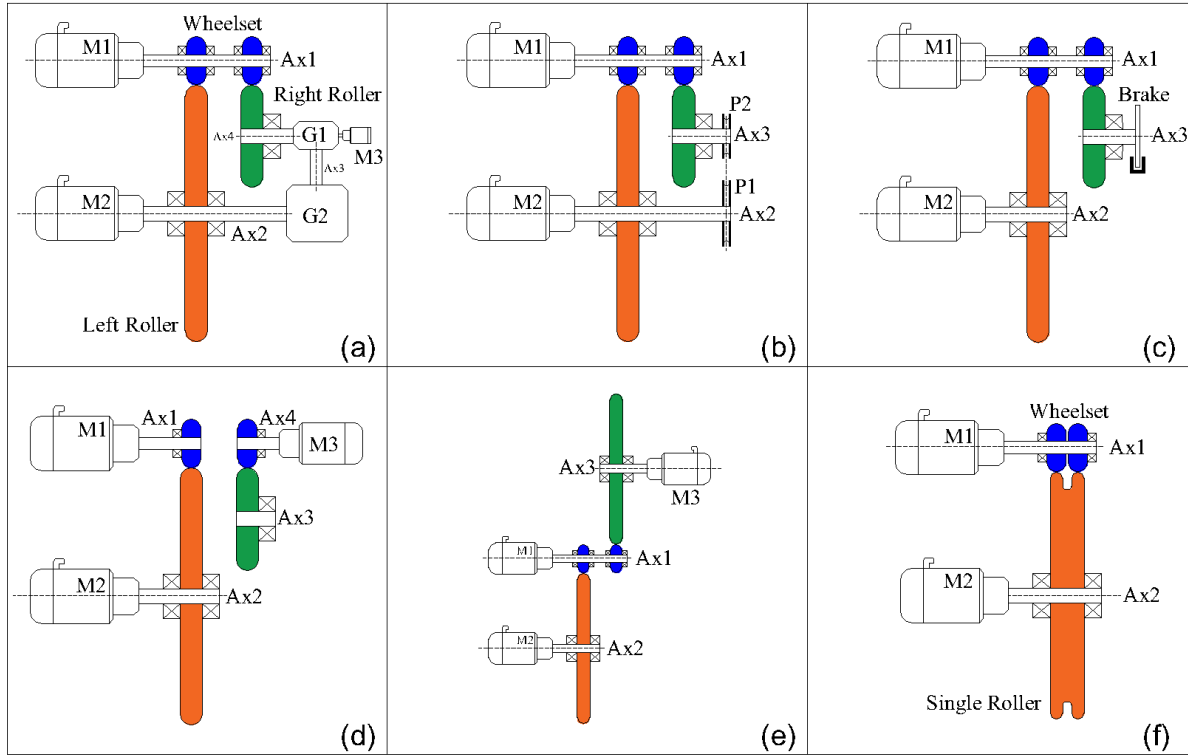


Figure 3: Schematic of six concepts considered for enabling single-axe testing. The top three concepts have a rigid wheelset and an independent, smaller right roller, and the bottom three concepts are alternative designs.

TABLE 1: Description, pros and cons of the possible concepts to emulate single axle testing in the VT-FRA Roller Rig

Concept	Description	Pros	Cons
(a)	Right roller is mechanically coupled to the left roller via a gearboxes G1 and G2; manual gear ratio control	Allows fine-tuning of the second roller speed	Multiple components, only large components are commercially available
(b)	Right roller is mechanically coupled to the left via a belt-pulley P1 and P2; the ratio is adjusted by changing the pulleys	Easy changing of the gear ratio by replacing pulleys	Low speed control accuracy, excessive compliance, and uncertainty about belt slip
(c)	Independent right roller; speed controlled by braking	Mechanically simple	Poor speed control, unstable under load; required additional actuator for braking
(d)	Idle right roller, an independent motor (M3) drives the second wheel	Measurement of the creep forces is facilitated	Poor wheelset emulation, and there is a large shaft deflection on the first axel, Ax1 (cantilever beam).

(e)	Two identical rollers; the second roller positioned above the wheelset	Symmetrical contact patches	High manufacturing and integration costs
(f)	Single roller with two rail profiles on the same shaft (Ax2)	Simplified structure	No independent creep control, limited DOF

Therefore, a modified concept based on the earlier designs was developed, as shown schematically in Figure 4. In this new approach, the second roller is powered by an independent, closed-loop electronic system, eliminating the need for purely mechanical power transmission as used in concepts 1 and 2 in Figure 3. This design offers the best balance between performance, reliability in speed control, and budget requirements.

The layout shown in Figure 4 comprises three independent drive motors (M1, M2, and M3), two independent rollers (referred to as left and right rollers), and a wheelset mounted on the upper shaft. Initially used to power a single wheel in the original setup, the first motor now powers and controls the wheelset's angular velocity. The second drivetrain, M2, is responsible for driving the left roller. Since they are part of the original setup, both motors M1 and M2 have the same power capacity (15.7kW) and features discussed in section 2. The new drivetrain, M3, powers the right roller independently from the other motors. This motor features a high-precision servo motor (equipped with a feedback unit) and a planetary gearbox.

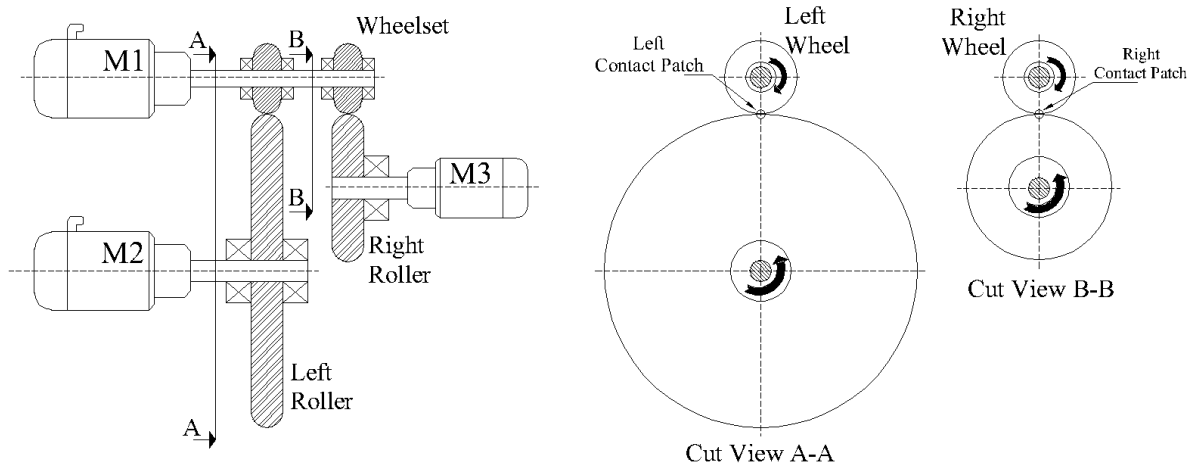


Figure 4: Schematic of the proposed configuration to enable single axle testing. The left side shows the main view, and the right side shows the cross-sectional view of each wheel-roller pair

The left roller has a diameter of 1120mm, while the right roller has a diameter of 406mm. The differences in diameter can be seen in the cross-sectional view in Figure 4, which shows two independent contact patches (CPs) referred to as the left and right CPs. The right roller features the standard US 136 rail profile to maintain consistency with the left (existing) roller. Additionally, the ratio between the right-hand wheel and the right roller is set to be approximately one-half.

Hardware selection

To enable multiple degrees of freedom for the wheelset in the proposed experimental setup in Figure 4, a new mechanism for the right-side roller must be developed. Figure 5 shows the schematic of this new mechanism, illustrating the independent roller's four degrees of freedom. Figure 5a depicts the right wheel

and roller in contact in the X-Z plane, highlighting the longitudinal and vertical displacements of the mechanism. The vertical displacement controls the right wheel load (i.e., the vertical load) through actuators L1 and L2. The longitudinal displacement is managed by a sliding platform that serves to fine-tune the roller position with respect to the right wheel. Figure 5b (Y-Z plane) shows the lateral displacement platform and the drive motor M3 responsible for controlling the roller's angular position.

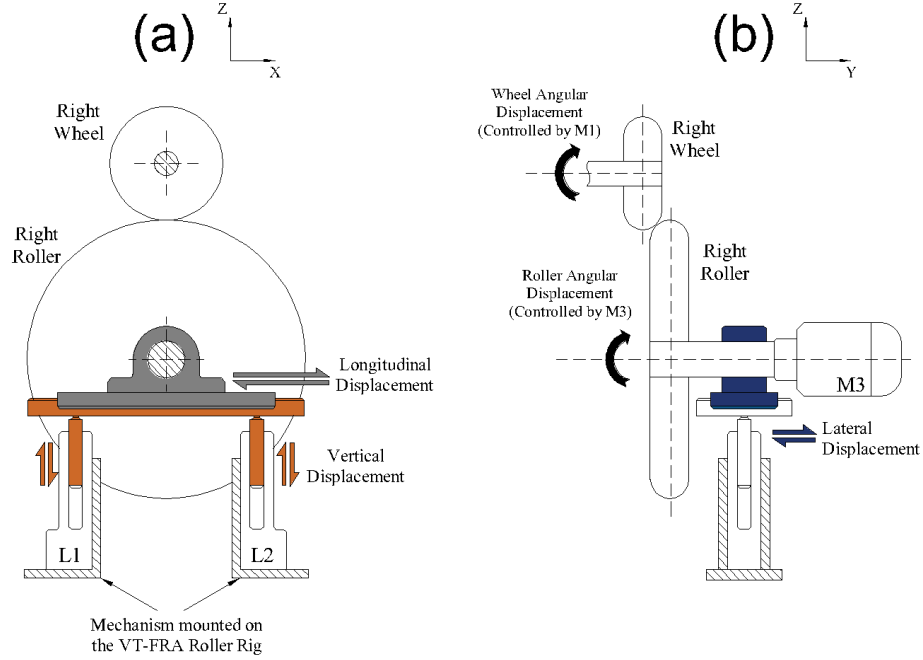


Figure 5: Schematic of the independent mechanism to be attached to the VT-FRA Roller Rig to control the angular velocity and positions of the right roller

To address the curve emulation challenges and to allow fine-tuning of the creep forces at the right contact patch, the drivetrain M3 must be appropriately selected. Therefore, a model of the longitudinal forces at this contact patch is developed.

The current VT-FRA Roller Rig was initially designed to apply a maximum vertical load of 10 kN to a single wheel [8]. Given this operational limit, the safe vertical load on the second contact patch of the wheelset is set to 5 kN. According to the Coulomb friction model [14], the maximum longitudinal (saturation) force that can be developed at the contact patch is directly proportional to the coefficient of friction (μ) and the normal force (F_z). Assuming a conservative friction coefficient of $\mu = 0.6$, the maximum expected longitudinal force is given by (2):

$$F_x = F_z * \mu = 5000 * 0.6 = 3000N \quad (2)$$

For small creepages, Kalker's linear model can be used [15]. In this regime, the longitudinal force developed at the wheel-rail interface is linearly proportional to the longitudinal creepage (γ_x) and the contact patch shape (ellipse with dimensions a and b) and the shear modulus of the material (G). Hence, for low creepages, the longitudinal force can be calculated by (3):

$$F_x = -Gabc_{11}\gamma_x \quad (3)$$

A model is created to simulate the torque in which the drivetrain M3 is expected to operate, and the results are shown in Figure 6. The left vertical axis shows the operational torque, and the right vertical axis shows the corresponding percentage creepage. The simulation spans from 0 to 60 seconds, with longitudinal creepage varying from -2% (braking) at the start to +2% (traction) at the end. The point at 30 seconds corresponds to pure rolling, where both the wheel and roller move at the same tangential velocity (longitudinal creepage, γ_x , is zero).

Using Kalker's and Coulomb's models, the operational torque is then calculated based on the second roller's diameter ($r_{roller} = 406mm$). The equation (4) describes the magnitude of the operational torque as a function of creepage:

$$T_z(\gamma_x) = \begin{cases} F_z \mu r_{roller}, & \text{for } \gamma_x < -0.36\% \text{ and } \gamma_x > +0.36\% \\ -Gabc_{11}\gamma_x r_{roller}, & \text{for } -0.36\% < \gamma_x < +0.36\% \end{cases} \quad (4)$$

Figure 6 shows that when the contact patch reaches saturation, represented by the flat segment of the blue curve, the maximum torque required is approximately 600 N.m. Therefore, the drivetrain is not expected to experience torque values exceeding this limit under normal operating conditions. According to the simulation, saturation occurs for creepages outside the $\pm 0.36\%$ creepage range. Additionally, inertial effects from the roller mass, motor, and gearbox are considered negligible when compared to the magnitude of the longitudinal forces generated by the bodies' relative slip.

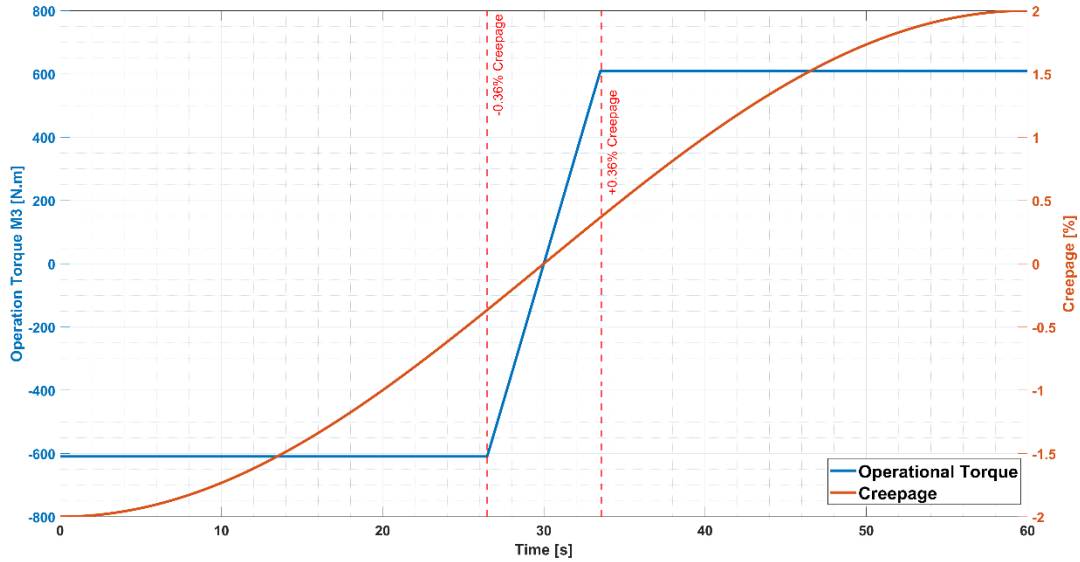


Figure 6: The right side vertical axis shows the creepage amplitude, and the left side vertical axis shows the operational torque required on the LSS of drivetrain M3 over time.

Besides the required torque that the drivetrain M3 will withstand, knowing the angular velocities at which the motor will operate is crucial. In curve emulation, for low-degree curves (e.g., 1-degree curves), the longitudinal velocities on each wheel will be close to the tangent track. Therefore, in the roller rig, the tangential velocities required on the right roller (governed by the drivetrain M3) will be close to the velocities on the left roller (governed by the drivetrain M2).

Figure 7 illustrates the linear relationship between the curve degree and the velocity differential ($\Delta\omega$) between the rollers, where the right roller represents the high rail and therefore spins faster. In the case of tangent track emulation, both rollers rotate at the same velocity (i.e., $\omega_r = \omega_l$), resulting in $\Delta\omega = 0$.

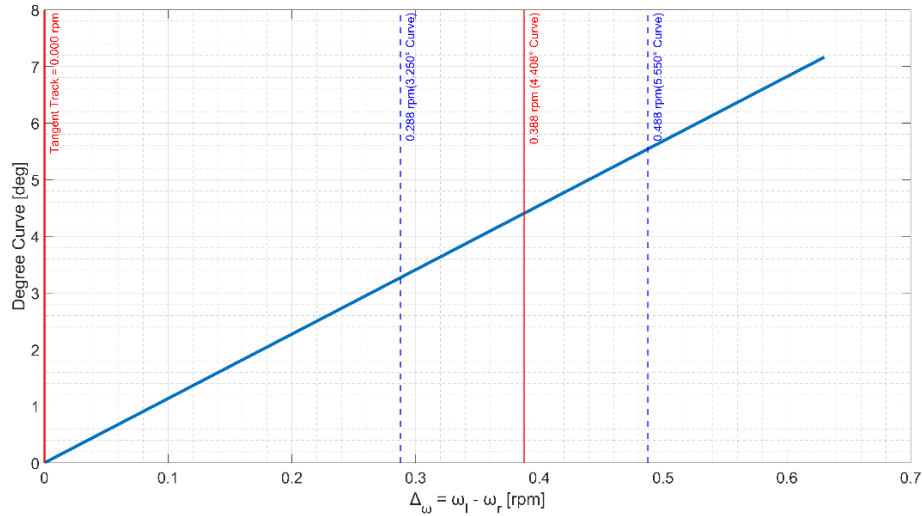


Figure 7: Degree curve as a function of the relative velocity on the right roller

The figure highlights that a velocity difference of 0.388 rpm is required to emulate a 4.4-degree (super-elevation) curve, which corresponds to negotiating a curve with a radius of 1300 feet. The vertical blue lines indicate two example curves of 3.25° (0.288 rpm) and 5.55° (0.488 rpm). This analysis reveals the sensitivity of curve emulation to roller velocity accuracy. For instance, if the 4.4-degree curve is the true value, a motor angular velocity accuracy of just ± 0.1 rpm in the right roller speed would translate into a curve emulation error of approximately 1.15 degrees (super-elevation), representing a significant deviation from the true value. Also, the drivetrain M3 must operate in a narrow angular velocity range. This range varies from the reference speed (equal to the left roller speed, at $\Delta\omega = 0$) up to approximately 0.7 rpm, corresponding to sharp curves. This analysis demonstrates that the M3 drive system must operate at a narrow resolution and achieve a level of accuracy as low as 0.01 rpm to maintain proper wheelset curve emulation.

To maintain consistency in maintenance and spare parts with the current VT-FRA Roller Rig, the same electric motor manufacturer, Kollmorgen, has been selected for the M3 drivetrain. The AKM65N is an independent three-phase permanent magnet servomotor controlled by a digital servo drive (Kollmorgen, S700 series). This servomotor can deliver a continuous 6.2 kW of mechanical power with a rated speed of 4000 rpm and a rated torque of 14.7 N.m. It has a high-resolution optical sine encoder (resolver) model DA, providing 2048 cycles per revolution [16]. Therefore, the resolver offers an accuracy of 0.333 arc-min (0.0055 degrees accuracy per turn). Since the motor output torque is relatively low, and at least 600 N-m is required, a gearhead is selected to amplify the 14.7 N-m motor's torque. The gearhead is positioned between the output of the electric motor (high-speed side - HSS) and the roller – the low-speed side (LSS). The selected planetary gearhead (model UltraTrue, UT014-50 [17]) features a gear ratio of 50:1, providing a nominal output torque of 671 N-m and a peak torque of 1360 N-m for momentary overload. The gearhead has a backlash of 5 arc-min (0.083° accuracy per turn) and high permissible radial and axial loads on the LSS of 22 and 20 kN, respectively. Since the roller is mounted in a cantilever configuration, the LSS will experience high forces and moments; therefore, high axial and radial permissible loads on the shaft are desirable.

With the M3's drivetrain selected, the velocity control accuracy can be estimated using the resolver accuracy, motor operational velocity, and the gearhead ratio. Since the encoder accuracy is rated as 0.0055 degrees, the percentage accuracy per revolution is calculated by equation (5):

$$\epsilon_{\%} = \frac{0.0055^{\circ}}{360^{\circ}} * 100 = \pm 0.0015\% \quad (5)$$

The encoder accuracy in the angular position measurement is proportional to the angular velocity accuracy in the LSS; therefore, the absolute accuracy in the velocity is given by (6)

$$\epsilon_{abs_{LSS}} = \epsilon_{\%} * n_n = \pm 0.0015\% * 4000rpm = \pm 0.061rpm \quad (6)$$

Since the gearhead increases the torque and decreases the velocity by a factor of 50, the angular velocity accuracy can be calculated by dividing the low-speed side accuracy ($\epsilon_{abs_{LSS}}$) by the gear ratio (N)

$$\epsilon_{abs_{HSS}} = \frac{\epsilon_{abs_{LSS}}}{N} = \pm \frac{0.061rpm}{50} = 0.0012rpm \quad (7)$$

According to the plot in Figure 7, such small orders of magnitude accuracy (i.e., 10^{-3}) can minimize the error to values as low as 0.011 degrees (super-elevation) curve (2.13ft radii). Also, the calculations above are conservative since the motor will operate at speeds lower than the motor's nominal capacity (i.e., 4000 rpm). Hence, the proposed hardware is suitable for curve emulation with enough accuracy.

Similarly to the roller's drivetrain, the wheel load must be controlled with equivalent accuracy. The synchronous linear actuators, represented by L1 and L2 in Figure 5, are used to control the roller's vertical displacement. The selected actuators (model PC40 from Thomson [18]) feature a 10 μ m accuracy that provides low backlash, thus increasing experimental repeatability. Beyond accuracy, the ball screw linear actuators offer a nominal thrust load of 6kN each, which is well beyond the design requirements. The actuator's cylinder is powered by an AC motor (model AKM33C from Kollmorgen) controlled by a servo driver series S700 from the same manufacturer. These linear actuators provide even higher accuracy than those used in the current VT-FRA Roller Rig (50 μ m).

The lateral and longitudinal displacements shown in Figure 5 are manually controlled by two independent platforms. Each platform features a rigid steel plate mounted on a caged ball linear block (provided by THK) that slides on a customized rail. The caged ball system offers low friction and backlash even under high loads [19]. The position of each platform is controlled by an ultra-precision lead screw/nut (with an accuracy of +/- 0.0006" per inch per turn) and a scaled handwheel for measuring the roller's position. Once the position is set both laterally and longitudinally, a locking mechanism comes into play, ensuring the roller remains in place.

The final CAD model including the proposed enhancements to the VT-FRA Roller Rig is illustrated in an exploded view in Figure 8. The additional wheel (right-side wheel) is installed on the existing upper shaft to create the wheelset. The independent mechanism to control the right-side roller positions and velocity is installed on the right side of the existing roller. The new mechanism frame is designed to be mounted in the available space and provides high structural stiffness under the dynamic loads. Also, following the initial design guidelines, the new system can be easily removed if single-wheel roller experiments are desired.

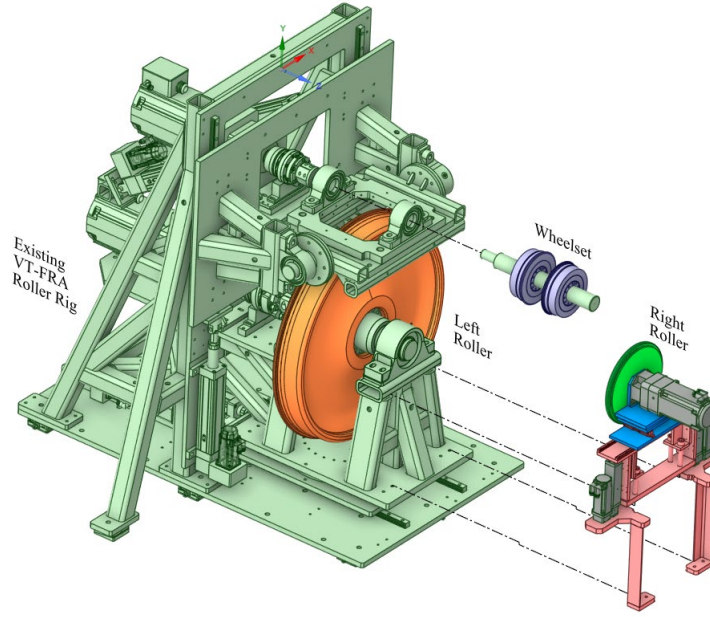


Figure 8: CAD model with the existing VT-FRA Roller rig on the left, the new wheelset, and the independent mechanism to control the right roller

NEW SETUP'S CAPABILITIES

The proposed configuration enables multiple experiments that mimic the field dynamics of a railcar wheelset. Figure 9 summarizes the enhancement of the existing setup: the left side shows the tangent track condition for mimicking traction and braking, and the right side shows the curving emulation capabilities.

Figure 9a depicts the tangential velocities on each wheel and the independent rollers. In this scenario, the tangential velocities of both rollers are kept the same to mimic the tangent track condition, i.e., $V_{left}^{roller} = V_{right}^{roller}$. The tangential velocities of the wheels are the same since the same shaft connects them. The magnitude of the wheel's velocities can be set to be different from the rollers to mimic traction ($V^{rollers} < V^{wheels}$), braking ($V^{rollers} > V^{wheels}$) or pure rolling ($V^{rollers} = V^{wheels}$). The traction studies can be performed in the centered position, i.e., the wheelset is centered with respect to the roller as shown in Figure 9b. In the case of lateral displacement of the rollers (keeping them aligned in the same horizontal plane), the wheelset unit must roll to maintain contact between the two contact patches due to the wheel's conicity. This condition is illustrated in Figure 9c and can be achieved due to the set of actuators that control the wheelset's cant angles. Therefore, traction studies can be performed with lateral displacement up to 25mm, with creepages varying from -2% to +2% and wheel load of 5kN on each contact patch.

The right side of Figure 9 illustrates the curving dynamics emulation capabilities of the proposed setup. Figure 9d presents the corresponding tangential velocities of the wheels (top) and rollers (bottom) during a simulated left-hand curve, where the right roller rotates faster than the left (i.e., $V_{left}^{roller} < V_{right}^{roller}$). Since both wheels are driven by a common drivetrain (M1), the tangential velocity at each wheel remains the same, as expected for a rigid axle wheelset. In the current configuration, actuators constrain the wheelset axially and laterally, meaning that no gravitational or lateral acceleration acts on the system. As a result, superelevation (cant angle) is not required to counteract centrifugal forces. Figure 9e shows the relative orientation of the wheelset and rollers during curve emulation. The schematic assumes balanced centrifugal forces, which justifies the absence of cant in this particular setup. However, in cases of superelevation-deficient curves, where centrifugal forces are unbalanced, this condition can still be replicated in the rig. By

precisely adjusting the lateral position of the rollers, additional lateral forces can be applied to the wheelset, effectively simulating unbalanced curving conditions.

Moreover, as shown in the top view in Figure 9f, the rollers can yaw independently from the wheelset while maintaining parallelism. This capability allows the rig to mimic wheelset yaw behavior during curve negotiation. Altogether, the proposed configuration enables high-accuracy curving emulation for a range of angles of attack (up to 5°) and curve degrees, from near-tangent tracks to sharp curves (e.g., 20°), thanks to the precision of the selected hardware.

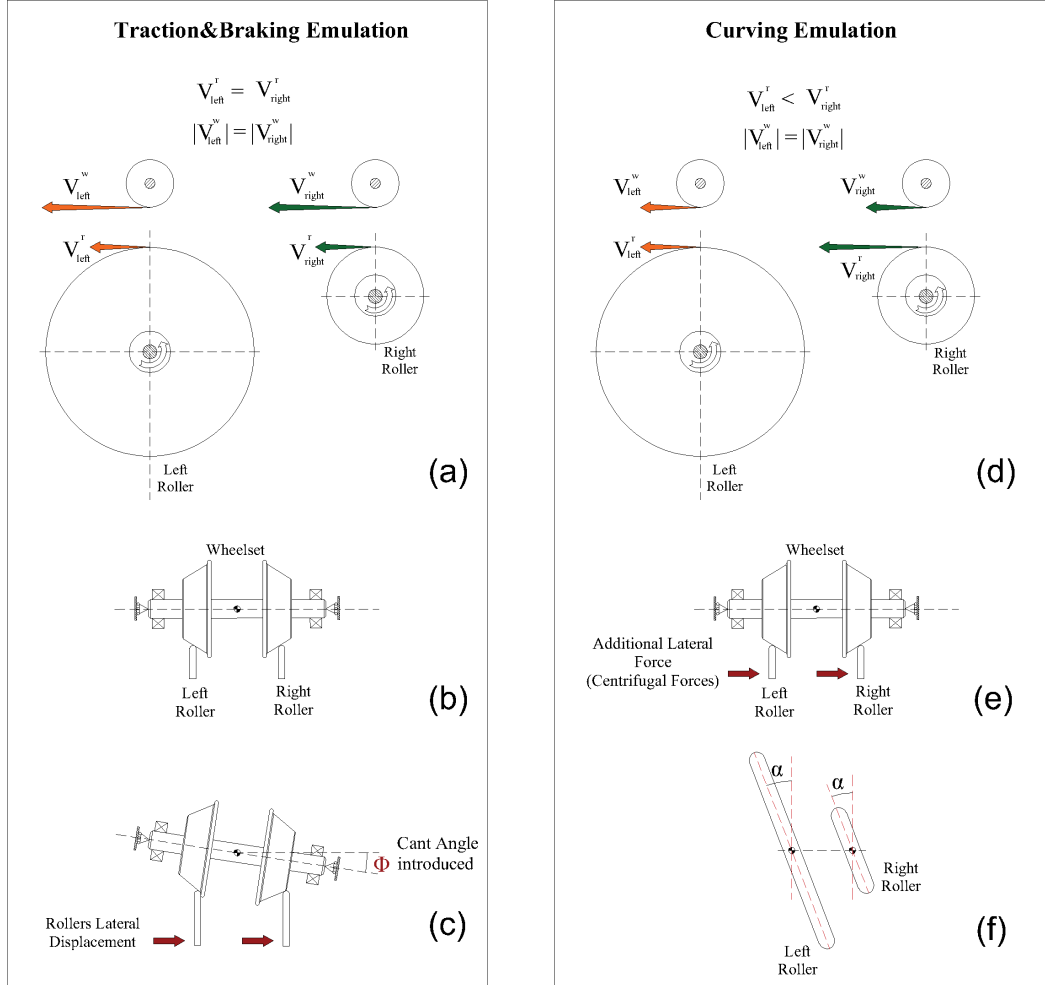


Figure 9 Capabilities of the new setup

The main limitation of the presented setup is the diameter of the second roller. While the system can accurately measure the longitudinal, lateral, and vertical contact forces, the contact shape plays an essential role in the creep forces at the wheel-rail interface. Even though the cross-sectional profile of each roller has the same dimensions (US 136 Rail scaled by four), the difference in roller diameters causes noticeable contact patch distortion [9]. Inevitably, left and right contact patches will have different distortions. Therefore, the pressure distribution will also be different. This characteristic could limit studies, focused on wear, since the right contact patch is expected to be shorter longitudinally and wider transversely compared to the left side. Consequently, different wear characteristics are expected on each contact patch. Even though the contact patch distortion will not affect the net forces on each wheel, direct comparisons between the two contact patches should be avoided.

CONCLUSIONS

This study presented the development of an enhanced VT-FRA Roller Rig configuration capable of performing single-axle (wheelset) testing to assess traction performance and curving dynamics under controlled laboratory conditions. The proposed design addresses a key gap in current experimental railway platforms: the lack of rigs capable of accurately controlling wheelset traction and replicating curving behavior.

A review of existing roller rigs demonstrated that while several systems are suitable for wear or bogie-level studies, none enable precise and independent control of the longitudinal velocities, creep forces, and wheelset degrees of freedom necessary for accurate curving emulation. To overcome these limitations, the enhanced configuration incorporates an additional roller and wheel, a set of synchronized actuators, and an independent drivetrain (M3) featuring a high-precision servo motor and planetary gearbox. This configuration supports emulating low to high-degree curves, with tight control over creepage, angle of attack, lateral displacement, and vertical wheel loads.

Simulation results confirm that the selected hardware meets the torque and velocity control requirements, achieving angular velocity resolution down to ± 0.0012 rpm, which translates to a curve emulation error of only ± 0.011 degrees. Similarly, the drivetrain's torque output, capped at 600 N-m, satisfies the expected force demands under both traction and braking conditions.

The new setup introduces robust measurement capabilities, leveraging the existing VT-FRA piezoelectric-based force measurement platforms. These features enable independent assessment of contact forces at both contact patches, a critical aspect for traction control studies and curve negotiation analysis.

Moreover, the platform can replicate field-like wheelset behaviors, including wheelset traction/braking, lateral displacement, and curve entry with yaw and compensation for superelevation, all within a laboratory environment. While the different roller diameters present contact patch distortion challenges, the system still allows accurate force quantification and dynamic testing. However, caution is advised for studies involving wear, as asymmetric contact patches are expected.

In conclusion, the upgraded VT-FRA Roller Rig significantly expands the existing single-wheel testing capabilities and helps bridge the gap between laboratory testing and revenue-service operation. It provides a valuable platform for validating traction strategies, creep force models, and wheelset-rail interaction mechanisms. Future work may include a deep analysis of the algorithm to calculate the creep forces from the piezoelectric sensor readings, prototyping the full system, and performing baseline tests with the new setup to evaluate its performance in replicating curve negotiation.

ACKNOWLEDGEMENT

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TABLE TITLES AND FIGURE CAPTIONS

TABLE 1: Description, pros and cons of the possible concepts to emulate single axle testing in the VT-FRA Roller Rig

Figure 1: Current VT-FRA Roller Rig setup. The left side displays the four degrees of freedom (a), and the sectional view on the right side (b) shows the single wheel and roller.

Figure 2: Requirements for curving emulation in a Roller Rig [1]. a) Set roller in the radial position; b) Left and Right roller with different angular velocities, and c) Roller unit tilting

Figure 3: Schematic of six concepts considered for enabling single-axle testing. The top three concepts have a rigid wheelset and an independent, smaller right roller, and the bottom three concepts are alternative designs.

Figure 4: Schematic of the proposed configuration to enable single axle testing. The left side shows the main view, and the right side shows the cross-sectional view of each wheel-roller pair

Figure 5: Schematic of the independent mechanism to be attached to the VT-FRA Roller Rig to control the angular velocity and positions of the right roller

Figure 6: The right side vertical axis shows the creepage amplitude, and the left side vertical axis shows the operational torque required on the LSS of drivetrain M3 over time

Figure 7: Degree curve as a function of the relative velocity on the right roller

Figure 8: CAD model with the existing VT-FRA Roller rig on the left, the new wheelset, and the independent mechanism to control the right roller

Figure 9 Capabilities of the new setup

Development of a Wheelset Roller Rig for Improving Rolling Stock Traction Efficiency

Giovanni Mantovani, Graduate Research Associate

Nikhil Kumar, Graduate Research Associate

Mehdi Ahmadian, J. Bernard Jones Chair and Director

Center for Vehicle Systems and Safety, Railway Technologies Laboratory

Virginia Tech

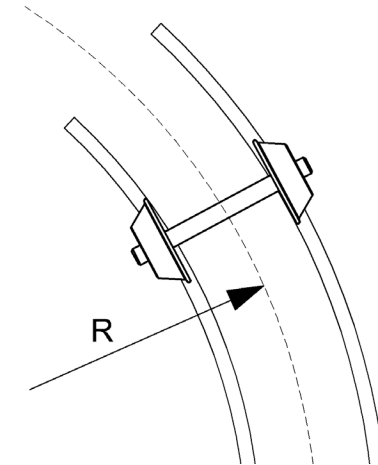
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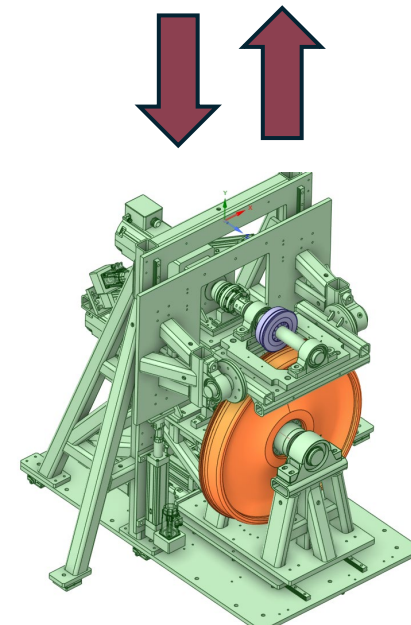
Indianapolis, IN

Motivation: Develop a wheelset-based test rig to enable accurately studying curving dynamics and ways of improving railroad safety

- **Roller rigs** are used for evaluating complex wheel-rail interactions (**WRI**) in a controlled and repeatable environment
- It is a scientific tool that provides results that can complement simulations and validate theories, thus providing suggestions to improve the standards
- A review of existing roller rigs highlights potential research gaps in the study of **curving dynamics** using wheelset-based roller rigs
- Research goals:
 - Develop a roller rig that enables testing a wheelset
 - Extending the existing Virginia Tech–Federal Railroad Administration (**VT-FRA**) Roller Rig
 - Address the existing gap in **creepage control** and wheelset **curving dynamics**



Wheelset Dynamics



Roller Rig Testing Dynamics



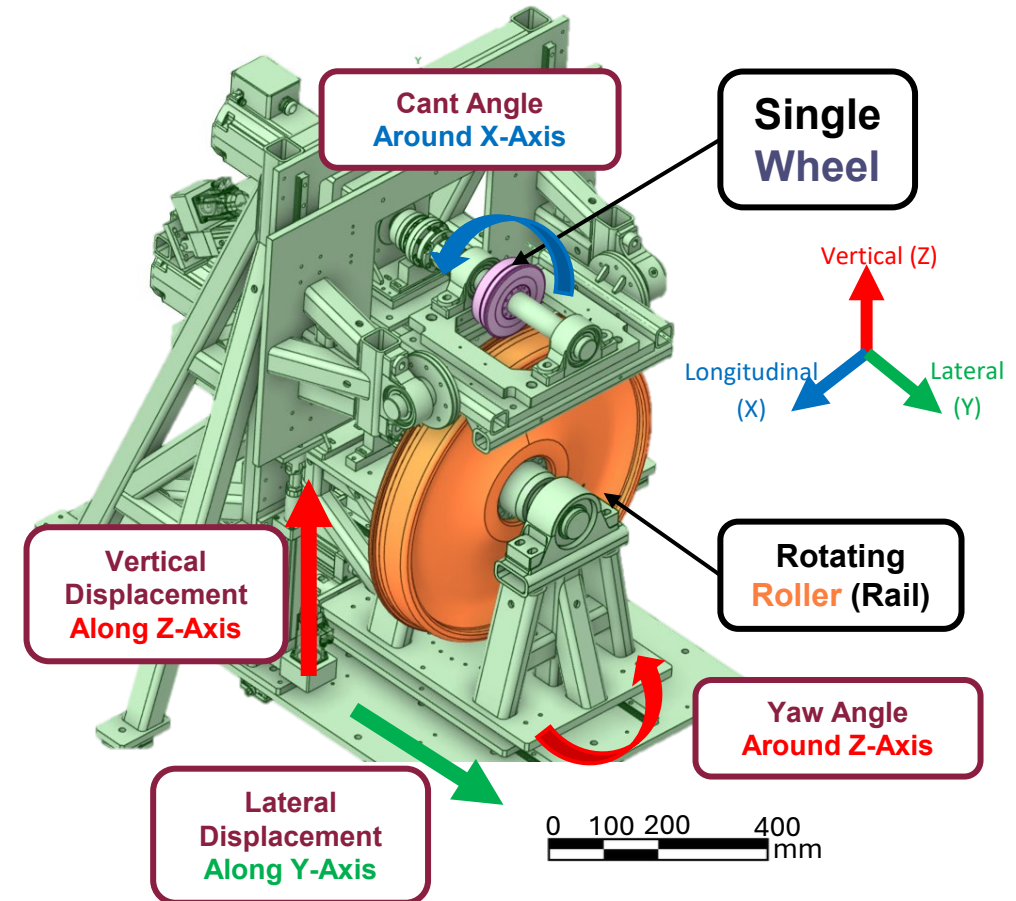
Background

VT-FRA Roller Rig and Wheelset Dynamics



VT-FRA Roller Rig Overview

- 1/4th Scaled Roller Rig used for WRI studies for a little more than a decade
- The forces developed at the contact patch can be measured with precision and high repeatability
- The goal is to add one more wheel and roller contact to the existing single-wheel rig to enable wheelset testing
 - Why? It enables studying curving dynamics

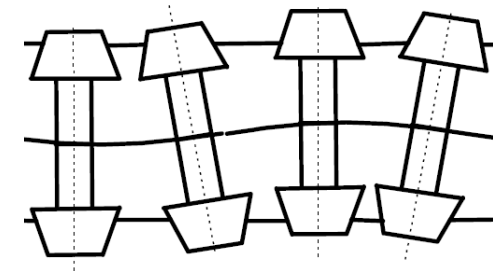
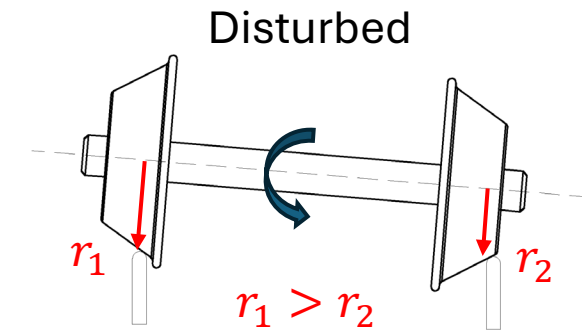
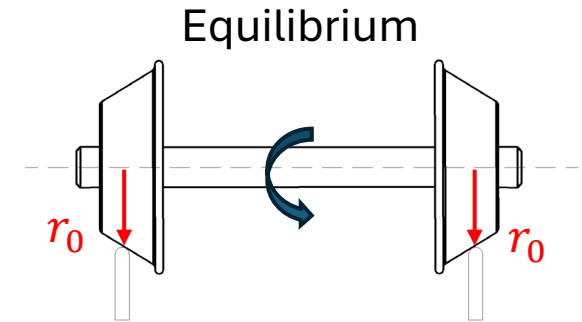


If you would like to read more about the capabilities of the VT-FRA Roller Rig, please scan this.



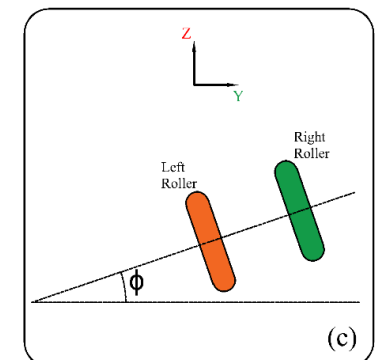
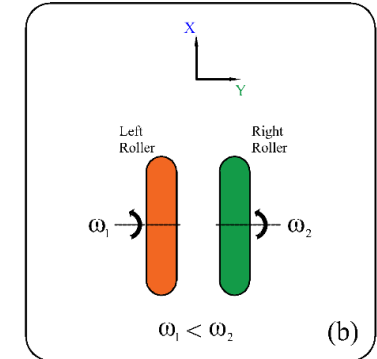
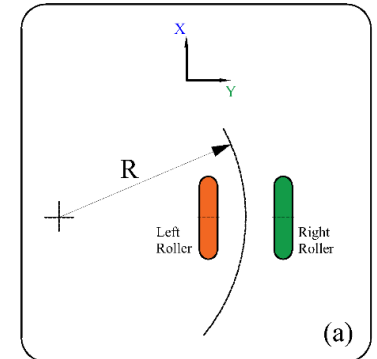
What happens when a wheelset negotiates a curved track?

- When a wheelset shifts laterally, the contact point at the WRI changes the rolling radius at each contact location
- The self-steering dynamics occur on both tangent and curved tracks
- The wheelset's unique geometry compensates for this disturbance
 - Wheels with symmetric **conicity** rotating at the **same angular speed**
- While the basic kinematic model explains how a wheelset is guided on a track, the contact-patch dynamics observes a far more complex behavior, particularly in curves
- In curves, the high-rail wheel moves forward with traction, while the low-rail wheel brakes or slips
 - Due to the radius differential of the two rails
- Roller rigs are essential tools for studying this complex phenomenon under controlled and repeatable conditions



Challenges of implementing the complex wheelset dynamics in the VT-FRA Roller Rig

- Three criteria that must be met:
 - a) Replicate of the wheelset angle of attack during curving
 - b) Simulate the differential speeds of the inner and outer wheels
 - This is done by setting the rollers with different velocities
 - c) Tilt the Roller unit to simulate curve superelevation
- Implementation Challenges:
 1. The wheel and roller are **mechanically constrained** by actuators
 2. **Limited space** restricts the addition of new instrumentation
 3. **Precise speed control** of the roller is required
 4. The system must withstand **high wheel loads**





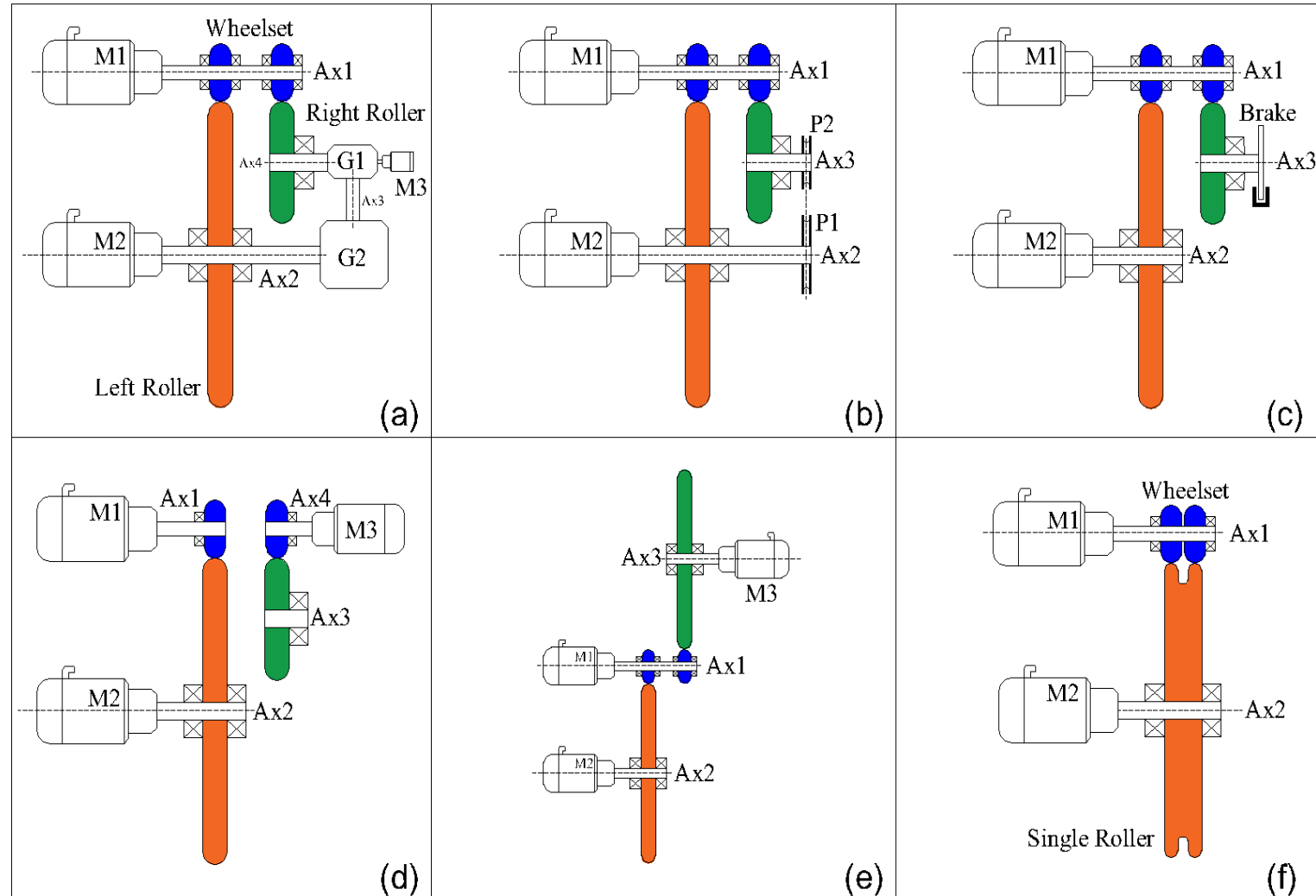
Design evaluation

- Our initial conceptual designs
- Capabilities of the final setup

Multiple configuration were assessed to enable wheelset testing with the VT-FRA Roller Rig

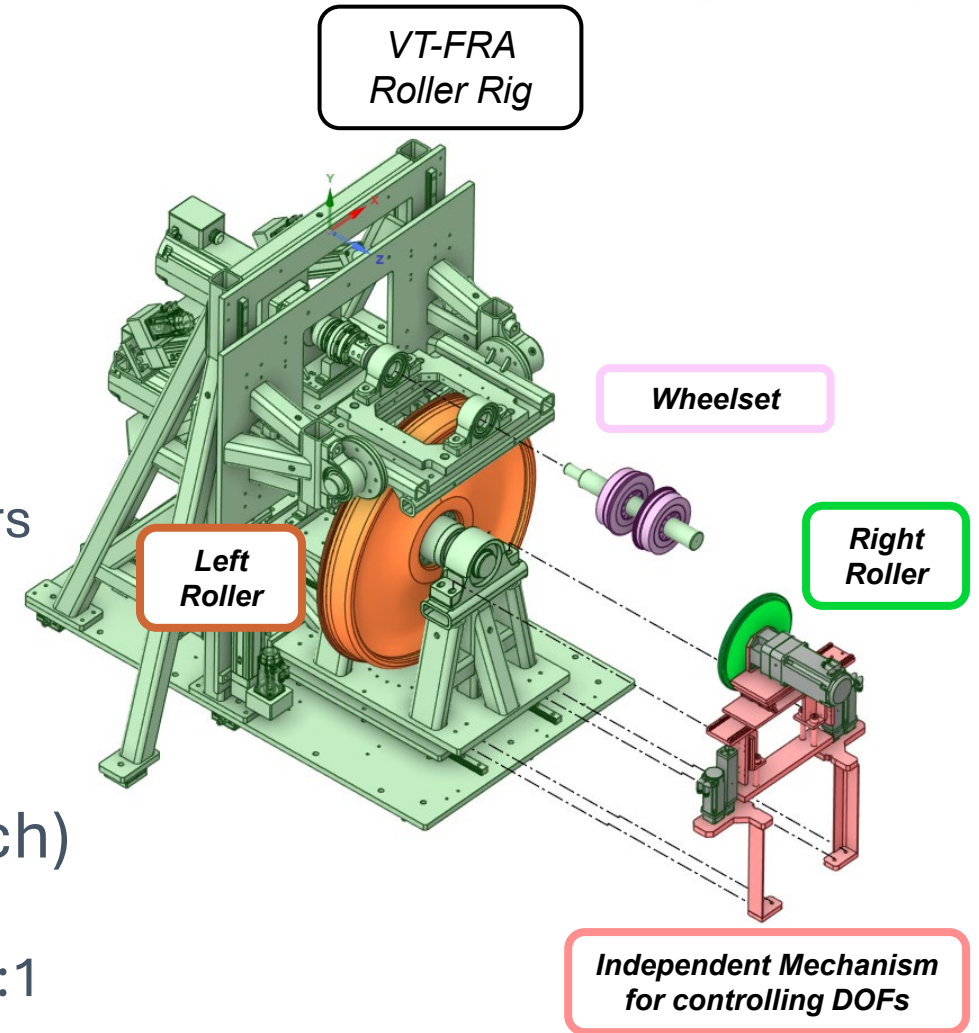
- Designs must comply with geometric constraints
- Designs must maintain the high precision of the existing single wheel-roller contact interface
- Each concept was evaluated and ranked for
 1. Performance
 2. Creepage control
 3. Position control
 4. Commissioning budget

Initial Concept Generation Ideas



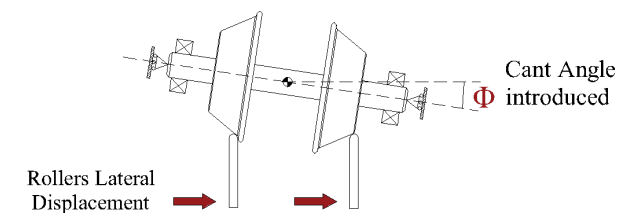
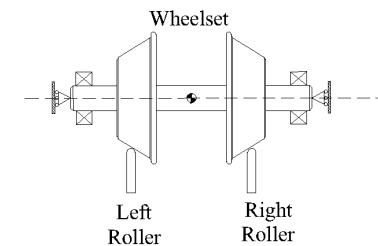
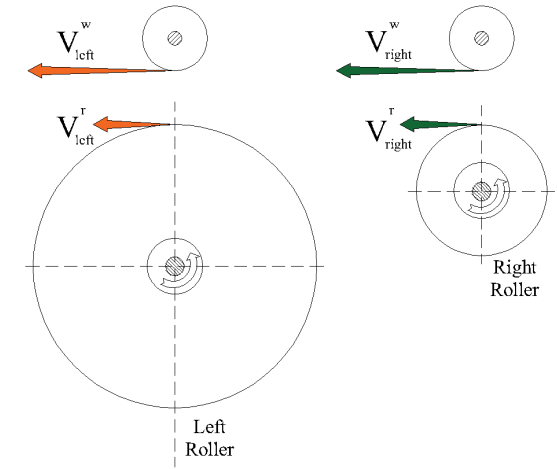
The final design uses the infrastructure of the VT-FRA Roller Rig to enable wheelset testing

- The redesign has a new roller of different size
 - Each roller is driven independently to achieve different velocities
- The new roller (right) is controlled by a new mechanism that allows
 - Precise control of the roller velocity
 - Adjusting vertical position (load) by linear actuators
 - Precisely positioning the wheel and roller position
- Contact forces can be measured at each contact patch independently
- Maximum wheel load of 5kN (per contact patch) can be achieved
 - Equal to 80kN for a full-scale wheel based on a 16:1 scaling factor



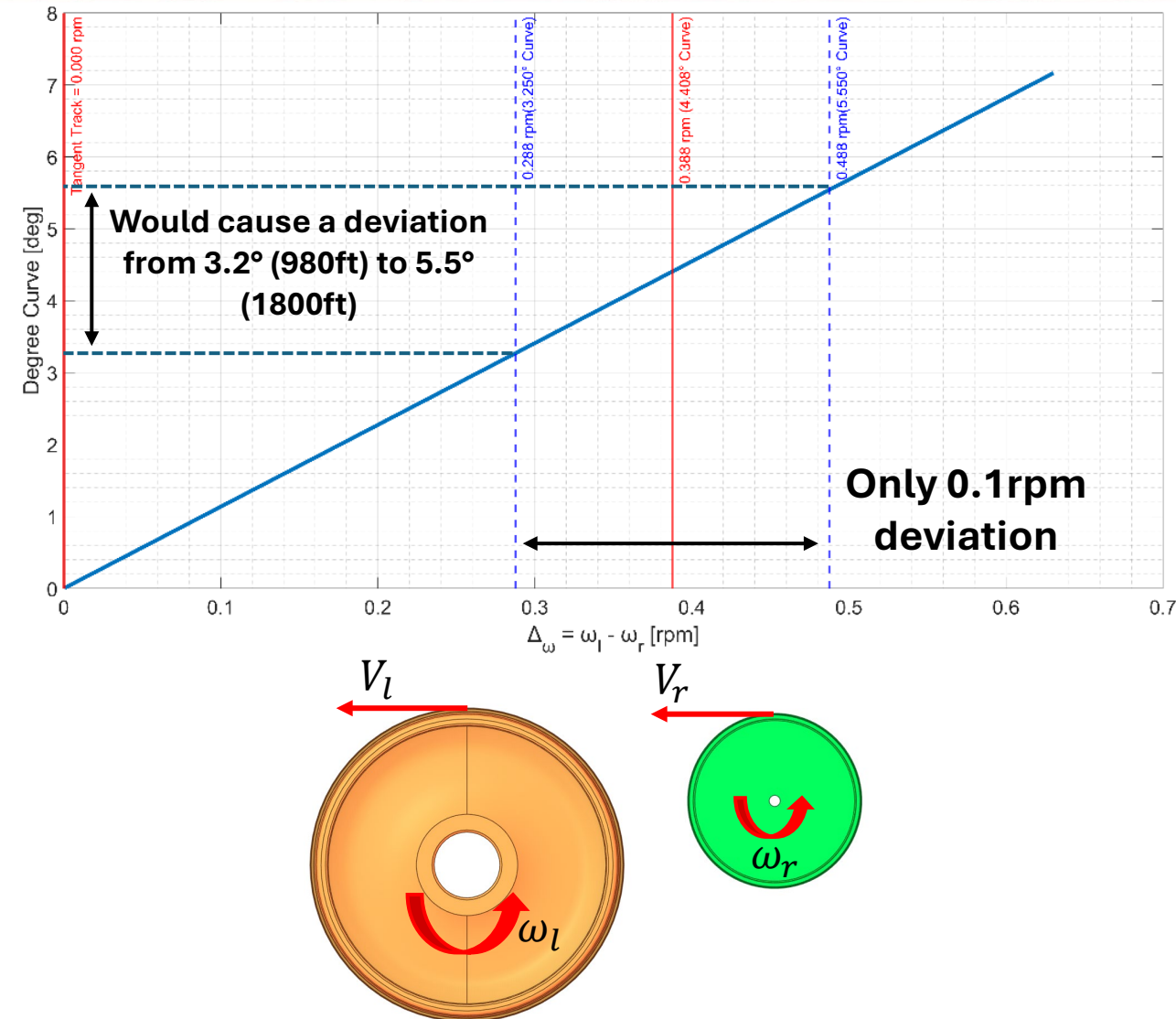
Wheelset traction (adhesion) and braking capabilities are achieved

- Separate motors drive the wheelset and rollers to
 - Independently control creepage at each contact patch
 - Emulate the creepage (speed) differential that occurs in a curve with high precision
- The rollers can be laterally displaced relative to the wheels
 - To emulate the lateral displacements of the contact patch on tangent tracks and in curves
- The wheelset can be held at cant angle relative to the rollers
 - To emulate the superelevation of different curves
- The longitudinal force differential that causes a yaw moment can be measured for the constrained wheelset using a novel force measurement arrangement



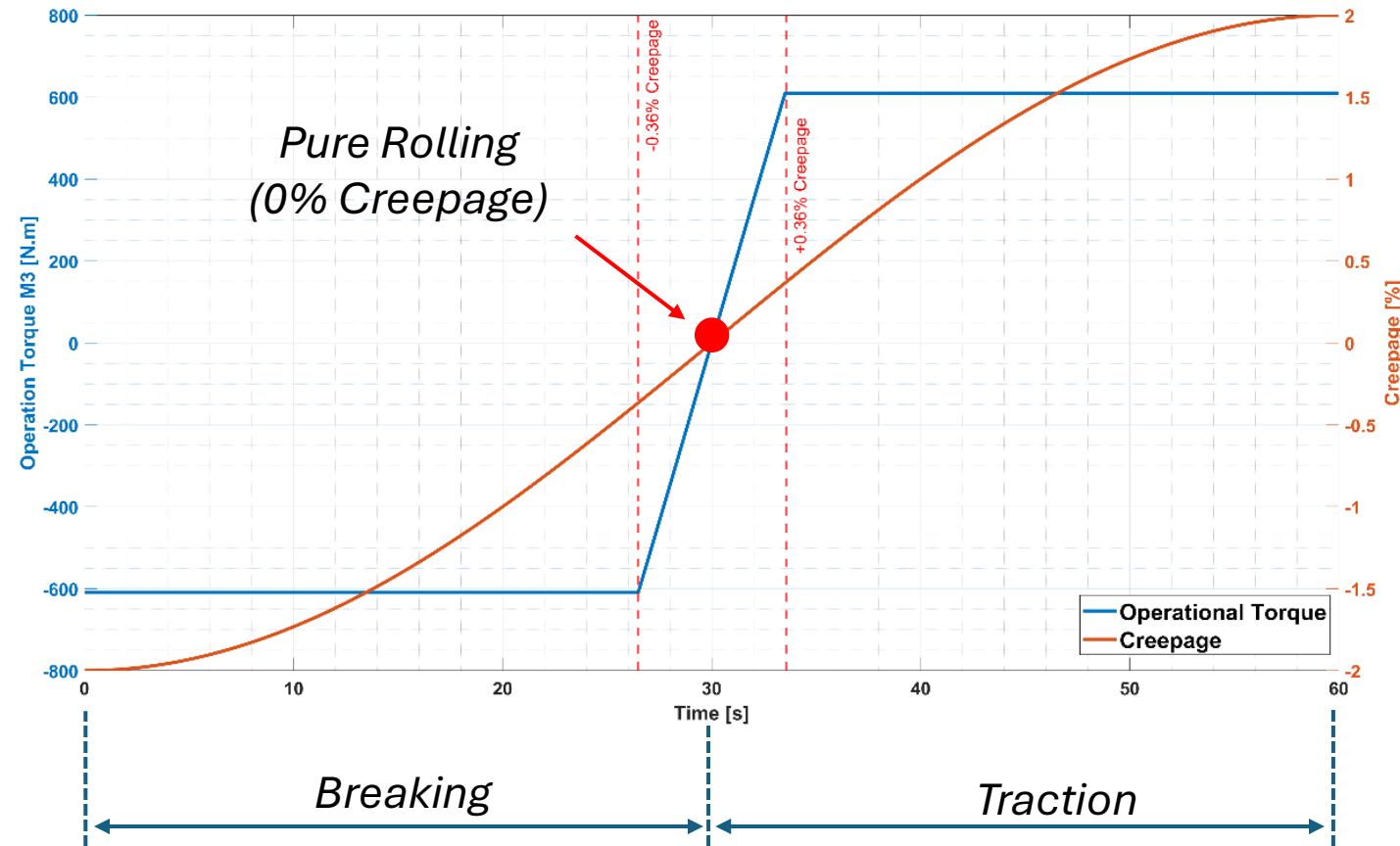
How crucial is roller speed control for emulating curves with precision?

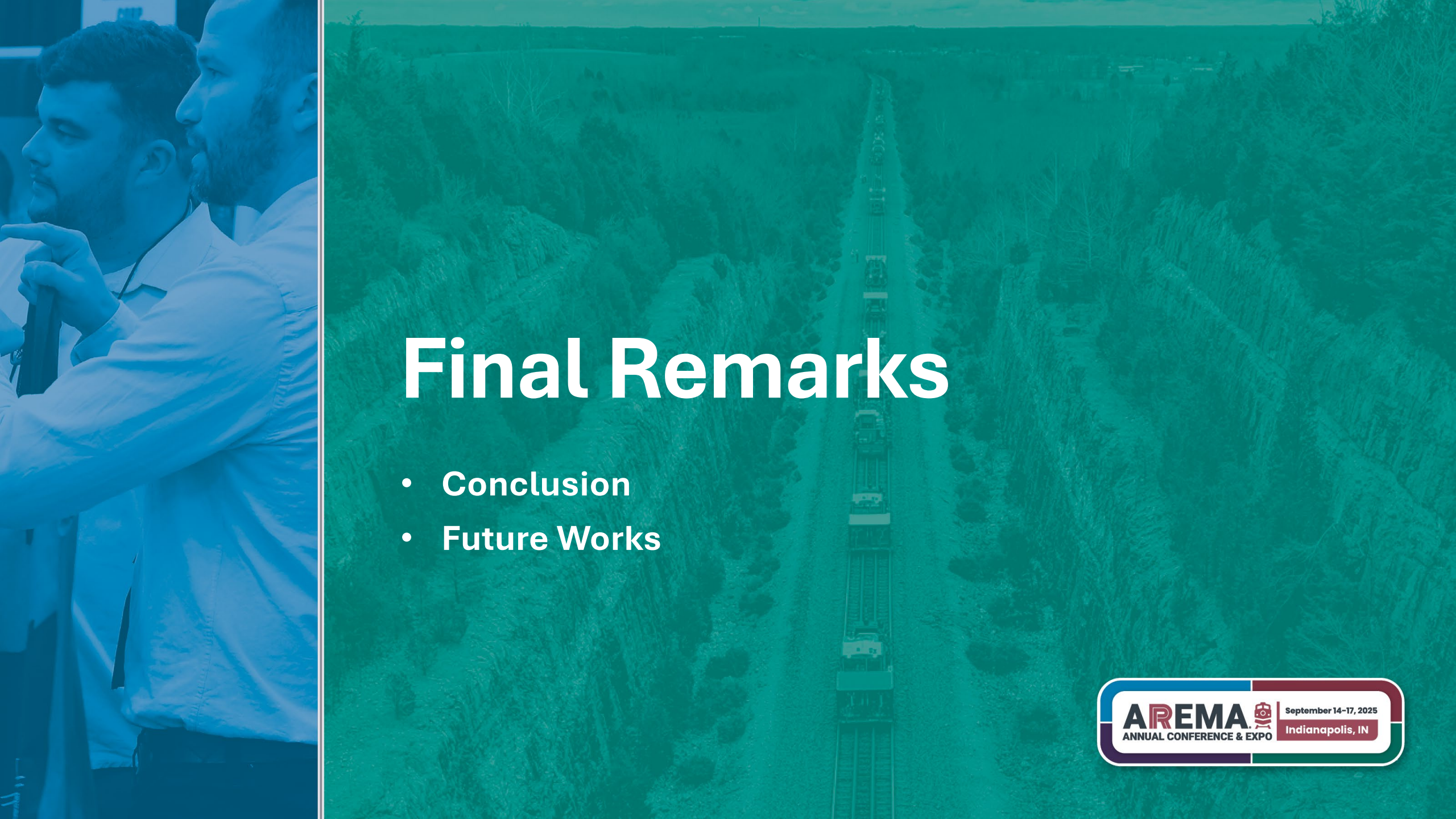
- The rollers can rotate independently
 - Equal roller speed emulates a **tangent** track
 - Different roller speeds emulate a **curved** track
 - The rollers' speed differential emulated the **degree of curvature**
- An analysis indicates that precise control of roller rotational speed is needed for emulating curves of different degrees
- Minor variations in roller speed (as small as <0.01 RPM) cause significant deviations to the curve degree
 - The motor control system can achieve a precision of 0.001rpm
 - This enables the emulation of varied curves on revenue service tracks



The new drivetrain can support high-creepage tests

- The right roller drivetrain is designed to accommodate a range of **braking** and **traction** conditions:
 - From -2% creepage (braking) to $+2\%$ creepage (traction)
- Longitudinal forces are modeled using Kalker's linear theory:
 - At small creepage values ($\pm 0.36\%$), the longitudinal creep forces vary linearly
 - Beyond this range, the forces saturate and no longer follow a linear relationship





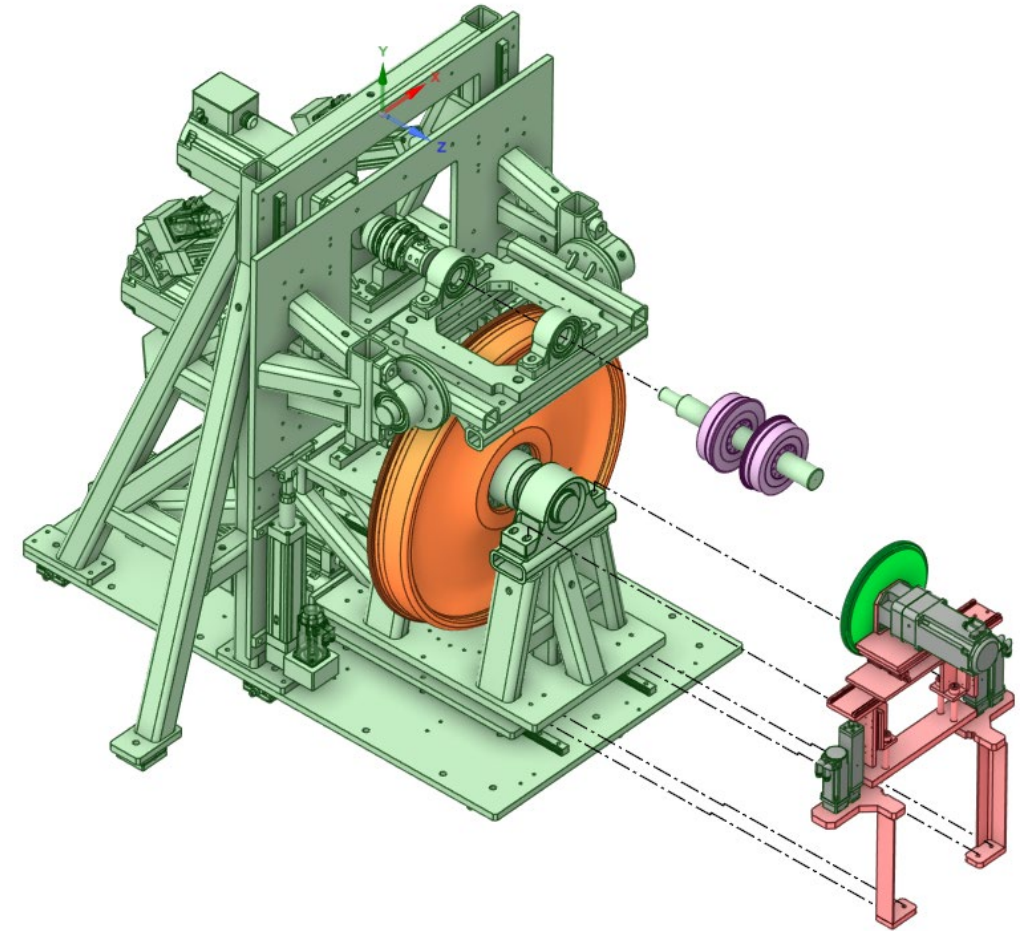
Final Remarks

- Conclusion
- Future Works

Conclusions

- The proposed design aims to successfully expand the current VT-FRA Roller Rig capabilities and enable testing with a single-axle
- The new rig addresses the industry's need for a high-precision system that allows scientific evaluation of a holistic understanding of dynamics that can improve railroad efficiency and safety
- The redesign
 1. Enables precise control of wheelset traction
 2. Accurately emulates wheelset under curving dynamics
- The selected hardware meets the stringent control requirements for rotational speed and torque precision
- Forces at each contact patch can be measured independently
- The rig's multiple degrees of freedom enable accurate control of yaw angle, lateral displacements, cant angle, and much more
- A calculated correction factor corrects the contact patch distortion caused by the different roller diameters

- Conduct a detailed analysis of the algorithm used to calculate creep forces
- Fabricate, calibrate, and commission the new rig by 2025
- Perform baseline experiments after commissioning, focusing on
 - Operational limits of traction and braking experiments
 - Curving emulation with multiple in various Angle-of-Attack (AoA) conditions
 - Establishing the contact patch distortion difference at the two wheel-rail contacts
 - And much more ...



Thank you!

- **Federal Railroad Administration (FRA) for providing funding for the project**
- **Ms. Alexandra D'Andrea and Dr. Abe Meddah of FRA for serving as program managers and for their technical suggestions to improve the project**
- **Mr. Bryan Marquis and Dr. Shushu Liu of VOLPE for many interesting technical discussions**
- **Mr. Ali Tajaddini (Retired FRA Program Manager) for envisioning the need for the VT-FRA Roller Rig and encouraging its expansion to a wheelset configuration**



How does the VT-FRA Roller Rig fulfill the requirements to emulate a curved track?

- The rollers are independently powered and can achieve different tangential velocities with high accuracy
 - This is kinematically equivalent to a curved track
- Emulation of the centrifugal forces:
 - Lateral displacement of the rollers creates additional lateral force on the wheelset
 - No cant angle is required in the Roller Rig to replicate curve negotiation
- Both rollers can be yawed to simulate the wheelset angle of attack during curving negotiation

